Homework 3 – STAT 6611

Due Friday, October 16 by midnight (submit in gradescope under 'Assignments')

1. For unit i, $i = 1, ..., n$, let y_i and x_i denote respectively the dependent variable and the known value of an independent variable. For a known value t, let z_i be defined as

$$z_i = \begin{cases} 0 & \text{if } x_i \leq t, \\ 1 & \text{otherwise,} \end{cases}$$

for $i = 1, ..., n$. Assume that the y_i's are uncorrelated with common variance σ^2, an unknown positive constant. Consider the (unconstrained) linear model

$$E(y_i) = \beta_0 + \beta_1 z_i + \beta_2 x_i + \beta_3 x_i z_i, \tag{1}$$

and the constrained linear model

$$E(y_i) = \gamma_0 + \gamma_1 z_i + \gamma_2 x_i + \gamma_3 x_i z_i, \text{ with the constraint } \gamma_1 + \gamma_3 t = 0. \tag{2}$$

You can assume that all β's and γ's are estimable with the available data.

   (a) For each of the two models, sketch in as much detail as possible what relationship the model proposes between the variables y and x. (A plot of y versus x, showing the general shape of the fitted model, could form an important part of your answer.)

   (b) For each of the following two statements, decide whether the statement is correct. Justify your answers.

   Statement 1: For model (1), the (ordinary) least squares estimator of β_2 depends only on those y_i's for which $x_i \leq t$.

   Statement 2: For model (2), the (constrained) least squares estimator of γ_2 depends only on those y_i's for which $x_i \leq t$.

For the following questions 2 and 3, let $G = \begin{bmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{bmatrix}$ be any generalized inverse of $\begin{bmatrix} X'X & A' \\ A & 0 \end{bmatrix}$ (the coefficient matrix of the constrained normal equations for Models

IIa–VIIa).

2. Consider Models IIa–VIIa. Let $\widehat{\underline{\beta}}$ be the first component of any solution to the constrained normal equations.

   (a) Show that $X\underline{\beta}$ has a unique least squares estimator $X\widehat{\underline{\beta}} = XG_{11}X'\underline{y} + XG_{12}\underline{b}$ and that the vector of residuals is $\widehat{\underline{\epsilon}} = (I - XG_{11}X')\underline{y} - XG_{12}\underline{b}$.

Use Theorem 2.14 and Lemma 2.13 to show (b) and (c).

   (b) $E(\widehat{\underline{\epsilon}}) = \underline{0}$ for all $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{b}$.

   (c) $XG_{11}X'$ and $I - XG_{11}X'$ are symmetric and idempotent matrices.

3. Take the model to be Model IIa and assume that $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ are both estimable. Let $\widehat{\underline{\beta}}$ be the first component of any solution to the constrained normal equations. Show that $\text{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'G_{11}\Lambda_2$.

4. Prove Theorem 2.18, the generalization of the Gauss-Markov theorem to Models IIa–VIIa.

5. Suppose that $\underline{y}$ is an $n \times 1$ observable random vector that follows the model

$$\underline{y} = X\underline{\beta} + Z\underline{b} + \underline{e},$$

where X is an $n \times p$ known matrix of rank p^*, Z is an $n \times q$ known matrix, and $\underline{\beta}$ is a $p \times 1$ vector of unknown parameters. Furthermore, suppose that $\underline{b}$ and $\underline{e}$ are independent random vectors with $\underline{b} \sim N(\underline{0}, \sigma^2 D)$ and $\underline{e} \sim N(\underline{0}, \sigma^2 I)$, where σ^2 is an unknown positive parameter, and D is a given positive definite matrix.

   (a) Show that this model is an Aitken model (Model III) with H nonsingular.

   (b) Give a BLUE of any estimable function $\underline{\lambda}'\underline{\beta}$. (Simplify your results to the greatest extent possible.)

   (c) Suppose that there exists a $p \times q$ matrix Q such that $Z = XQ$. Give a simple expression for a BLUE of any estimable function $\underline{\lambda}'\underline{\beta}$ and explain your reasoning. The expression that you give should NOT involve D.

6. Consider the following model for a balanced two-way classification:

$$y_{ij} = \mu + \tau_i + \gamma_j + \epsilon_{ij} \quad (i = 1, ..., t, \quad j = 1, ..., c);$$

where $\mu, \tau_1, ..., \tau_t, \ \gamma_1, ..., \gamma_c$ are unknown parameters; where $\epsilon_{11}, ..., \epsilon_{tc}$ are random errors, with $E(\epsilon_{ij}) = 0$, $\text{Var}(\epsilon_{ij}) = \sigma^2$, $\text{corr}(\epsilon_{ij}, \epsilon_{ij'}) = \rho$ for $j' \neq j$, and $\text{corr}(\epsilon_{ij}, \epsilon_{i'j'}) = 0$ for $i \neq i'$; and where σ^2 and ρ are unknown parameters satisfying $\sigma^2 > 0$ and $\rho > -1/(c-1)$ with $c > 1$. Establish whether or not the variance of the SLSE of every estimable function is uniformly less than or equal to that of any other linear unbiased estimator of that estimable function.

Hint: Writing $\underline{y} = (y_{11}, ..., y_{1c}, ..., y_{t1}, ..., y_{tc})'$, $\underline{\epsilon} = (\epsilon_{11}, ..., \epsilon_{1c}, ..., \epsilon_{t1}, ..., \epsilon_{tc})'$, and $\underline{\beta} = (\mu, \tau_1, ..., \tau_t, \gamma_1, ..., \gamma_c)'$, then the model matrix X (which is $(tc) \times (1 + t + c)$) and the variance matrix (which is $(tc) \times (tc)$) can be expressed as

$$X = \begin{bmatrix} \underline{1} & \underline{1} & \underline{0} & \cdots & \underline{0} & I \\ \underline{1} & \underline{0} & \underline{1} & \cdots & \underline{0} & I \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \underline{1} & \underline{0} & \underline{0} & \cdots & \underline{1} & I \end{bmatrix}. \qquad V = \text{Var}\,(\underline{\epsilon}) = \sigma^2 \begin{bmatrix} A & & & \\ & A & & \\ & & \ddots & \\ & & & A \end{bmatrix}.$$

Above $\underline{1} = (1, ..., 1)'$ and $\underline{0} = (0, ..., 0)'$ are $c \times 1$ column vectors, while I be a $c \times c$ identity matrix and $A = (1 - \rho)I + \rho\underline{1}\,\underline{1}'$.